

TONIGHT

Luck, or **signal**?

Last week every answer ended with "it looks different". Tonight we find out whether it is, with one picture instead of a formula.

Data Analysis & Machine Learning for Business

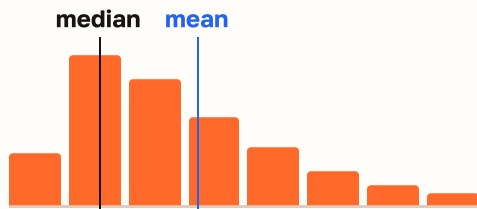
Andrey Shumilin

Three pictures we already have.

01

Histogram

A column is a shape, not a number.
The width of the shape is the randomness, and it is why one number always tells a partial truth.



02

Median and quartiles

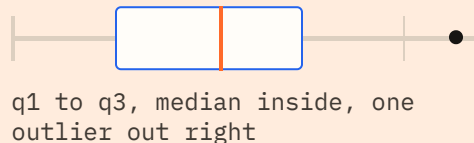
Sort the column and cut it. The median is the middle value, q1 and q3 are the quarter marks, p90 is a promise: nine of ten tasks within 28 minutes.



03

Boxplot

All of them in one picture: q1, median, q3, whiskers at 1.5 IQR. A dot past the whisker is data, not an error. And two of them sit side by side.



All three describe one group. Tonight we put two of them next to each other and ask whether the gap is real.

Five beats, and then your checklist.

- | | | |
|----|---|----------------------|
| 01 | Chance alone produces differences | one coin flip |
| 02 | The luck machine, and the p-value read off it | 10,000 shuffles |
| 03 | Choosing a test without a PhD | five routes |
| 04 | Two real cases, one of them lies to us first | after the break |
| 05 | Three traps that look like findings | and how to spot them |

It ends with the checklist that **HW1 Part 2 rewards**.

Chance alone produces differences.

THE ONLY HONEST QUESTION

If these two groups really were one and the same, how often would chance hand me a gap this big?

Build the world where they are the same, count how often it beats you, and that share is the answer. The simulation has a name: the permutation test. Nothing is measured in it. Only the labels move.

step 1

Take the 925 task paces of sectors 5761 and 5130, and pool them.



step 2

Deal them back at random into piles of 304 and 621. A coin flip decides everything.



step 3

-2.14 picks per hour between the piles

Bigger than the gap we came to explain, and pointing the other way.

Vote now. Sector 5761 runs 1.8 picks per hour ahead. Real difference, or luck?

Demo 01

Shuffle the labels ten thousand times and watch what luck alone can produce.

THE PICTURE OF THE EVENING

p = 0.66 6,588 of 10,000

Two thirds of pure-luck worlds beat our observation of +1.80 picks per hour.

- 1 The per-task table in one expression, on picks_per_hour, plus two asserts
- 2 Sectors 5761 against 5130: medians 135.3 and 133.5.
Vote: real or luck?
- 3 New tool, toy first: six numbers, two piles, and a gap of 38 out of thin air
- 4 **10,000 shuffles, the histogram, the red line.** Card 1 on the second screen
- 5 The other pair, 5402 against 5400: gap +36.5, zero shuffles of ten thousand
- 6 The textbook words afterwards, and the named tests in one line each

A big p is a failure to find a difference, not proof of sameness.

THE DEFINITION YOU JUST WATCHED

p = the share of luck-worlds at least as extreme as your observation.

01

~~The probability
my hypothesis is
true~~

It is a property of the luck-worlds, not of your belief.

02

~~The probability
the result is due
to chance~~

It assumes chance and asks how often chance does this.

03

~~The importance of
the effect~~

**p measures surprise,
not size.**

04

~~p = 0.06 means
nothing is there~~

It means this evidence is not enough.

05

~~p < 0.05 means
the finding
matters~~

**With 150,000 rows,
a 2% effect clears
the bar.**

0.05 is a convention, not a law of nature. It is a threshold you agree on **before** looking, priced by what a false alarm costs this decision.

A named test is the same shuffle, solved in advance.

Two groups of numbers

times, amounts, durations

→ `stats.ttest_ind(a, b)`

Start here. It compares the means, and at our sample sizes it holds up on a skewed column too.

and the column has a long tail

a few values hundreds of times
the typical

→ `stats.mannwhitneyu(a, b)`

A **second opinion** beside it, for when the mean stops describing the typical case.

Shares of yes or no counts

error rates, conversion, defect
share

→ `stats.chi2_contingency()`

Feed it **raw counts**, one row per group. Never percentages.

A number against a number

task size against pace, weight
against time

→ **cut it into groups first**

No test here takes two number columns. **Defend where you cut.**

Unsure, or a home-made metric

medians, ratios, p90

→ **permutation test**

Shuffle the labels 10,000 times and count. Always legal.

And no test at all is legal before the earlier question is answered: **does my metric measure the thing I mean?**

Demo 02

Two hypotheses born from the process. The second one turns on us.

THE UNIT CHANGES TONIGHT

Whole tasks run on `picks_per_hour`, bigger is faster. Single picks run on **seconds per pick**, where smaller is faster.

Case 1	Shelf levels. Hypothesis and what would change my mind, written first
	The U-shape: 6.6 on level 1, 5.4 in the middle, 8.0 to 8.2 up the ladder
	Shapes first, then two tests: t-test $1.5e-14$ against ranks $1.6e-195$
AI	The lazy prompt against the prompt that carries the data
	What else differs? Weight is flat, and light items keep the effect
Case 2	SmallParts. The obvious metric says ratio 0.95. Let it hang
	Card 4, rebuild per cell visit: 17.8% against 26.9%, and the add-back check

No test rescues a metric built on a **misunderstood process**.

Demo 03

Three traps, all respectable-looking.
Each one runs on data whose truth we
set ourselves.

Trap 1 Regression to the mean

"We took the three slowest aside, and next week they were faster."
Twenty invented workers whose pace never changes: the slowest
five gain 10.1% in 86 weeks out of 100. Our own pickers, 118 pairs,
slope 0.75.

Trap 2 Multiple comparisons

Twenty meaningless splits, two of them clear 0.05 by design. Then
18 real categories on 172,016 attempts: 13 significant, 11 survive
Bonferroni. At big n , significance is cheap.

Trap 3 Significant but tiny

Night against day on 145,904 picks: t -test 0.38, ranks $4.1e-05$, and
the mean and the median point opposite ways. When two honest
tests disagree, that is the finding.

Decisions run on effect sizes and mechanisms. **p only guards the gate.**

Six lines, each one earned tonight.

01 Hypothesis written before looking, and what evidence would change your mind.

02 The metric matches the process. SmallParts.

03 The test matches the data. A long tail gets a rank test beside the t-test.

04 p-value and effect size reported together. The 18 categories.

05 Confounders checked. Weight on the floor level.

06 One look: the horizon, the slices and the test decided in advance.

HW1 does not require a statistical test.

A clear split, a chart and a defended verdict is already a complete Part 2, exactly as the README said. An honest test is one of the three ways to earn the last two points.

Six ideas to keep.

01 Luck impersonates signal, so simulate the luck.

02 p is the share of luck-worlds beating you. Not truth, not importance.

03 The extremes of any ranking bounce back on their own.

04 t -test first, a rank test beside it when the tail is long.

05 Metric before test: know the process first.

06 Effect size decides, p only guards the gate. One look, decided in advance.

NEXT WEDNESDAY

We stop explaining and start predicting: machine learning on tables. New dataset, and for once it is home turf. Financial ratios of real companies, and the question is who goes bankrupt. HW1 is due before that class and HW2 goes out the same evening, so front-load.